

NumericalAlgebra 4

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PROBLEM I Use the Taylor series to show that the approximation

$$\frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) (x_i, y_j) \approx \frac{a_{i+\frac{1}{2},j}(u_{i+1,j} - u_{i,j}) - a_{i-\frac{1}{2},j}(u_{i,j} - u_{i-1,j})}{h_x^2}$$

is second-order accurate, provided that

$$a \frac{\partial u}{\partial x} \in C^3, \quad a \frac{\partial^3 u}{\partial x^3} \in C^1.$$

That is, show that the absolute value of the difference between the right-hand side and left-hand side is bounded by

$$\frac{h_x^2}{24} \left[\max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial^3}{\partial x^3} \left(a \frac{\partial u}{\partial x} \right) \right| + \max_{x \in [x_{i-1}, x_{i+1}]} \left| \frac{\partial}{\partial x} \left(a \frac{\partial^3 u}{\partial x^3} \right) \right| \right].$$